SYNTHETIC TEST DOCUMENT - Fictional Polytechnic University

Bachelor of Science in Mechanical Engineering - Course Descriptions

MATH 101 Calculus I 4
Limits and continuity, differentiation of elementary functions, the definite integral, the fundamental theorem of calculus, techniques of integration, area and volume.

MATH 201 Linear Algebra and Differential Equations 4
Matrix algebra, determinants, eigenvalues, first and second order linear ordinary differential equations, Laplace transforms, vector calculus, line and surface integrals.

PHYS 101 Physics for Engineers 4
Kinematics in one and two dimensions, Newton's laws, conservation of energy and momentum, waves, electric fields, magnetism, simple DC and AC circuits.

CHEM 105 Engineering Chemistry 3
Stoichiometry, chemical equilibrium, acid-base equilibria, thermochemistry, electrochemistry, chemical kinetics and bonding.

CS 110 Programming for Engineers 3
Structured programming in C and Matlab, functions, arrays, text files, algorithm design and testing, numerical methods including root finding and numerical integration.

STAT 220 Probability and Statistics for Engineers 3
Probability, random variables, discrete and continuous probability distributions, sampling, estimation, hypothesis testing, regression.

MECH 210 Statics and Dynamics 4
Equilibrium of particles and rigid bodies, trusses, friction, kinematics and kinetics of particles and rigid bodies, work, energy, impulse and momentum.

MECH 220 Mechanics of Materials 3
Stress and strain, axial loading, torsion of shafts, bending of beams, shear stress, deflection of beams, buckling of columns, combined stresses.

MECH 230 Fluid Mechanics 3
Fluid statics, conservation of mass, momentum and energy, Bernoulli equation, dimensional analysis, viscous flow in pipes, boundary layers.

MECH 240 Engineering Thermodynamics 3
Properties of pure substances, first and second laws of thermodynamics, entropy, ideal gas, power and refrigeration cycles.

MECH 250 Materials Science 3
Atomic structure, crystal structures, phase diagrams, mechanical properties of metals, polymers, ceramics and composites, heat treatment.

MECH 260 Engineering Graphics and CAD 3
Orthographic projection, sectional views, dimensioning and tolerancing, assembly drawings, computer aided design.

MECH 310 Applied Thermodynamics and Heat Transfer 4
Rankine, Otto, Diesel and Brayton cycles, vapour compression refrigeration, psychrometry; steady and transient conduction, natural and forced convection, radiation heat transfer, heat exchangers.

MECH 320 Kinematics and Dynamics of Machines 4
Kinematic analysis of planar mechanisms, static and dynamic force analysis, cams and gears, balancing of rotating machinery, free and forced vibration of single and multi degree of freedom systems, vibration isolation.

MECH 330 System Dynamics and Control 3
Mathematical models of mechanical, electrical and hydraulic systems, transfer functions, block diagrams, step and frequency response, Bode diagrams, stability criteria, proportional integral derivative control.

MECH 340 Machine Design 4

Design of machine elements, stress concentration, fatigue under variable loads, bolted and welded joints, springs, shafts, bearings, clutches and brakes; manufacturing processes, joining, material removal and process selection.

EE 250 Electrical and Electronic Circuits 3

Analogue and digital semiconductor devices, diodes, transistors, operational amplifiers, digital logic, electric machines and power.

MECH 410 Fluid Machinery 3

Pumps, fans, compressors and turbines, similarity laws, cavitation, pipeline systems and turbomachinery performance.

MECH 420 Finite Element Methods 3

Finite element formulation for bars, beams, trusses and plane stress problems, stiffness matrices, mesh refinement and computer implementation.

MECH 430 Robotics 3

Robot kinematics, forward and inverse kinematics, manipulator dynamics, trajectory planning, sensors and actuators.

HUM 200 Technical Communication 2

Report writing, oral presentations and professional correspondence.